

[Aim: 100/100 in Maths]

आरंभ

ARITHMETIC PROGRESSION

A.P

Lecture #2

ATK^o:-

fixed no. $\Rightarrow$ Common difference (d)

[अंतराल - फिक्स्ड]

AP की पहचान? $\rightarrow$ all c.d. must be same

General term of AP: $a, a+d, a+2d, a+3d, \dots, a+(n-1)d$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $a_1 \quad a_2 \quad a_3 \quad a_4 \quad \dots \quad a_n$

nth term of AP

$$a_n = a + (n-1)d$$

#LP: In which of the following situations, does the list of numbers involved make an arithmetic progression, and why?

✓ i. The taxi fare after each km when the fare is ₹15 for the first km and ₹8 for each additional km.

FARE: ₹15, ₹23, ₹31, ₹39, ₹47, ...

a_1 a_2 a_3 a_4 a_5

+8 +8 +8 +8

$$d_1 = a_2 - a_1 \Rightarrow 23 - 15 \Rightarrow 8$$

$$d_2 = a_3 - a_2 \Rightarrow 31 - 23 \Rightarrow 8$$

$$d_3 = a_4 - a_3 \Rightarrow 39 - 31 \Rightarrow 8$$

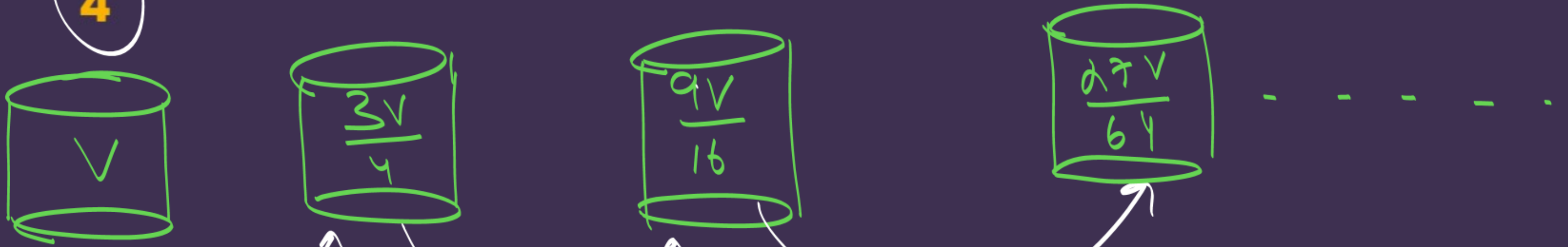
$$d_4 = a_5 - a_4 \Rightarrow 47 - 39 \Rightarrow 8$$

Since all c.d. are same.

∴ It is AP.

ii. The amount of air present in a cylinder, when a vacuum pump removes $\frac{1}{4}$ of the air remaining in the cylinder at a time.

4



$$\rightarrow V - \frac{1}{4}V$$

$$\rightarrow \frac{3V}{4}$$

$$\rightarrow \frac{3V}{4} - \frac{1}{4}\left(\frac{3V}{4}\right)$$

$$\rightarrow \frac{3V}{4} \left(1 - \frac{1}{4}\right) \rightarrow \frac{3V}{4} \times \frac{3}{4} = \frac{9V}{16}$$

$$\rightarrow \frac{9V}{16} - \frac{1}{4}\left(\frac{9V}{16}\right)$$

$$\rightarrow \frac{9V}{16} \left(1 - \frac{1}{4}\right)$$

$$\rightarrow \frac{9V}{16} \times \frac{3}{4} \rightarrow \frac{27V}{64}$$

$$\begin{matrix} V & \frac{3V}{4} & \frac{9V}{16} & \frac{27V}{64} & \dots & AP?? \\ a_1 & a_2 & a_3 & a_4 & & \end{matrix}$$

$$\begin{aligned} d_1 &= a_2 - a_1 \Rightarrow \frac{3V}{4} - V \Rightarrow \frac{3V - 4V}{4} \Rightarrow \left(\frac{-V}{4} \right) \\ d_2 &= a_3 - a_2 \Rightarrow \frac{9V}{16} - \frac{3V}{4} \Rightarrow \frac{9V - 12V}{16} \Rightarrow \frac{-3V}{16} \end{aligned} \quad \left. \vphantom{\begin{aligned} d_1 &= a_2 - a_1 \Rightarrow \frac{3V}{4} - V \Rightarrow \frac{3V - 4V}{4} \Rightarrow \left(\frac{-V}{4} \right) \\ d_2 &= a_3 - a_2 \Rightarrow \frac{9V}{16} - \frac{3V}{4} \Rightarrow \frac{9V - 12V}{16} \Rightarrow \frac{-3V}{16} \end{aligned}} \right\} \rightarrow d_1 \neq d_2$$

Since c.d are
not same

Not AP !!

iii. The cost of digging a well after every metre of digging, when it costs ₹150 for the first metre and rises by ₹50 for each subsequent metre.

₹150, ₹200, ₹250, ₹300 - - -
 a_1 a_2 a_3 a_4

$$d_1 = a_2 - a_1 \Rightarrow 200 - 150 = 50$$

$$d_2 = a_3 - a_2 \Rightarrow 250 - 200 = 50$$

$$d_3 = a_4 - a_3 = 300 - 250 = 50$$

are all same

∴ Yes

iv. The amount of money in the account every year, when ₹10000 is deposited at compound interest at 8% per annum.

Initial $\Rightarrow$ ₹ 10000 a_1

After 1 yr $\Rightarrow$ $10000 \left(1 + \frac{8}{100}\right)^1$ a_2

After 2 yr $\Rightarrow$ $10000 \left(1 + \frac{8}{100}\right)^2$ a_3

After 3 yr $\Rightarrow$ $10000 \left(1 + \frac{8}{100}\right)^3$ a_4

$$d_1 = a_2 - a_1$$

$$= 10000 \left(1 + \frac{8}{100}\right) - 10000$$

$$= 10000 \left[1 + \frac{8}{100} - 1\right]$$

$$= 10000 \left(\frac{8}{100}\right) \Rightarrow \boxed{800} = d_1$$

$$d_2 = a_3 - a_2$$

$$= 10000 \left(1 + \frac{8}{100}\right)^2 - 10000 \left(1 + \frac{8}{100}\right)^1$$

$$\Rightarrow 10000 \left(1 + \frac{8}{100}\right) \left[1 + \frac{8}{100} - 1\right] \Rightarrow 10000 \left(1 + \frac{8}{100}\right) \left(\frac{8}{100}\right)$$

$$CI = P \left(1 + \frac{R}{100}\right)^T$$

$$\begin{aligned} &800 \left(1 + \frac{8}{100}\right) \\ &= 800 + 800 \times \frac{8}{100} \\ &= 800 + 64 \end{aligned}$$

$$d_2 = \boxed{864}$$

Since $d_1 \neq d_2$

Not AP

#LP: The first term of an A.P. is -7 and the common difference 5.
Find its 18th term and the general term.

$$a_1 = -7$$

$$d = 5$$

$$a_n$$

To find
i) $a_{18} = ??$

$$a_n = a + (n-1)d$$

$$\begin{aligned} \text{i) } a_{18} &= a + (18-1)d \\ &= -7 + (17)(5) \\ &= -7 + 85 \end{aligned}$$

$$a_{18} = 78$$

$$\begin{aligned} \text{ii) } a_n &= a + (n-1)d \\ a_n &= -7 + (n-1)5 \end{aligned}$$

$$a_n = -7 + 5n - 5$$

$$a_n = 5n - 12$$

#LP: The 6th and 17th terms of an A.P. are 19 and 41 respectively, find the 40th term.

$$\begin{array}{l} a_6 = 19 \\ a_{17} = 41 \\ \boxed{a + 5d = 19} \text{ --- (I)} \\ \boxed{a + 16d = 41} \text{ --- (II)} \end{array}$$
$$\begin{array}{r} a + 5d = 19 \\ - (a + 16d = 41) \\ \hline -11d = -22 \\ d = 2 \end{array}$$

put $d = 2$ in (I)

$$a + 5(2) = 19$$
$$a + 10 = 19$$
$$\boxed{a = 9}$$

To find

$$a_{40} = a + 39d$$

$$a_{40} = 9 + 39(2)$$

$$a_{40} = 9 + 78$$

$$\boxed{a_{40} = 87}$$

#LP: If 9th term of an A.P. is zero, prove that its 29th term is double the 19th term.

$$\begin{aligned} a_9 &= 0 \\ \wedge \\ a + 8d &= 0 \\ a &= -8d \quad \text{--- (1)} \end{aligned}$$

To prove $a_{29} = 2a_{19}$

LHS $a_{29} = a + 28d$
 $= -8d + 28d$

LHS $= 20d$ ✓

RHS

$$\begin{aligned} 2a_{19} &\Rightarrow 2[a + 18d] \\ &\Rightarrow 2[-8d + 18d] \\ &\Rightarrow 2[10d] \end{aligned}$$

RHS $\Rightarrow 20d$ ✓

Since
LHS = RHS

HP

Quad Module

$$2P(I) \rightarrow Q1 - 15$$

$$2P(II) \rightarrow Q1 \rightarrow 10$$

THANK YOU COODIESS !!

